

DIABETIC RETINOPATHY SCREENING REPORT

Smart India Hackathon 2026 – Problem Statement 26038

Image ID: sample_undus
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1. CLINICAL SUMMARY

AI Diagnosis: Moderate NPDR

Severity Level (0-4): 2

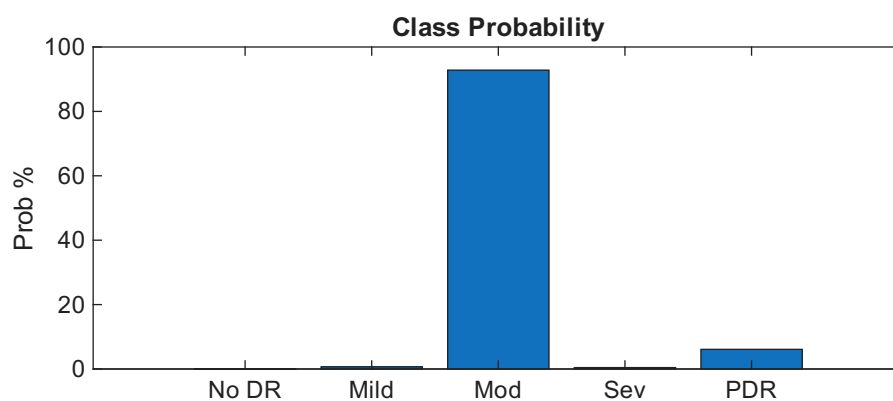
Confidence: 92.8%

URGENT REFERRAL – Tele-consult within 7 days

Microaneurysms (MA): 24

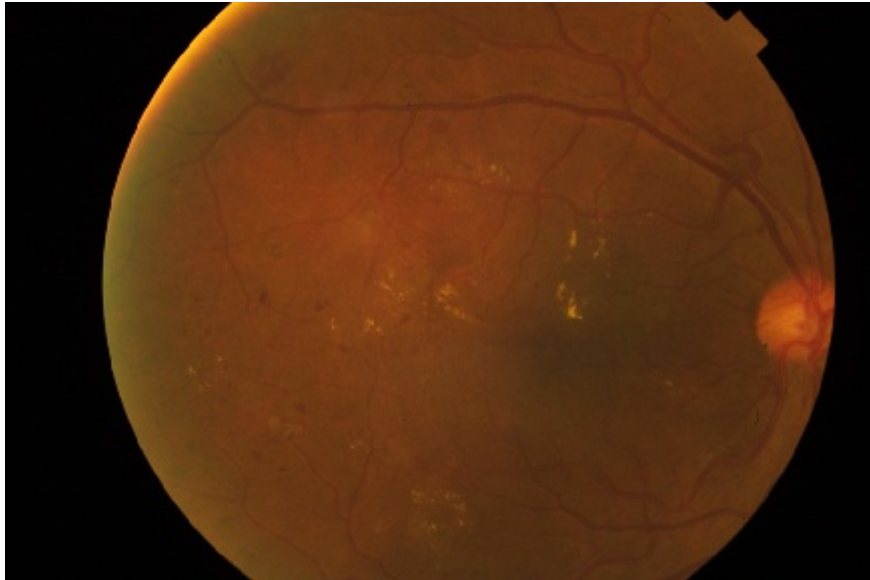
Hard Exudates (EX): 1

Hemorrhages (HE): 0

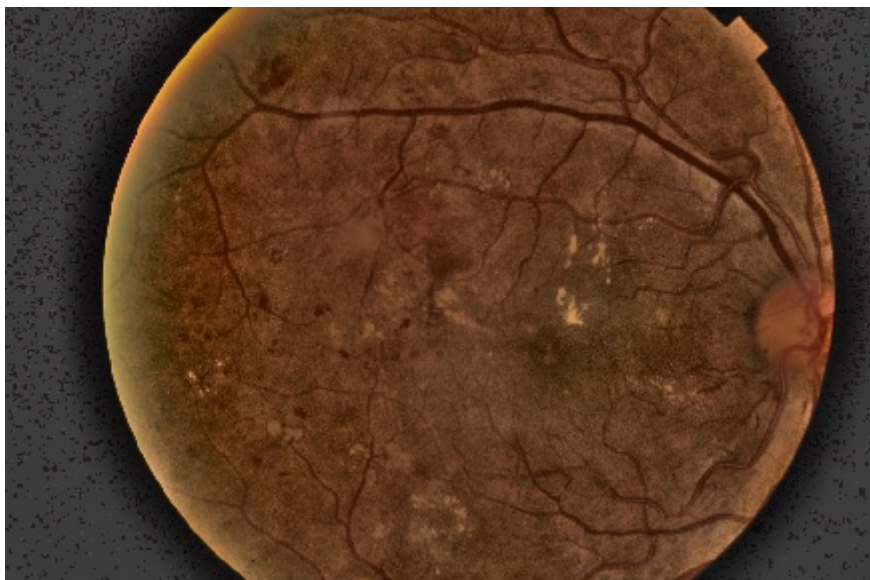


2. ORIGINAL & ENHANCED FUNDUS IMAGES

Original Fundus Image

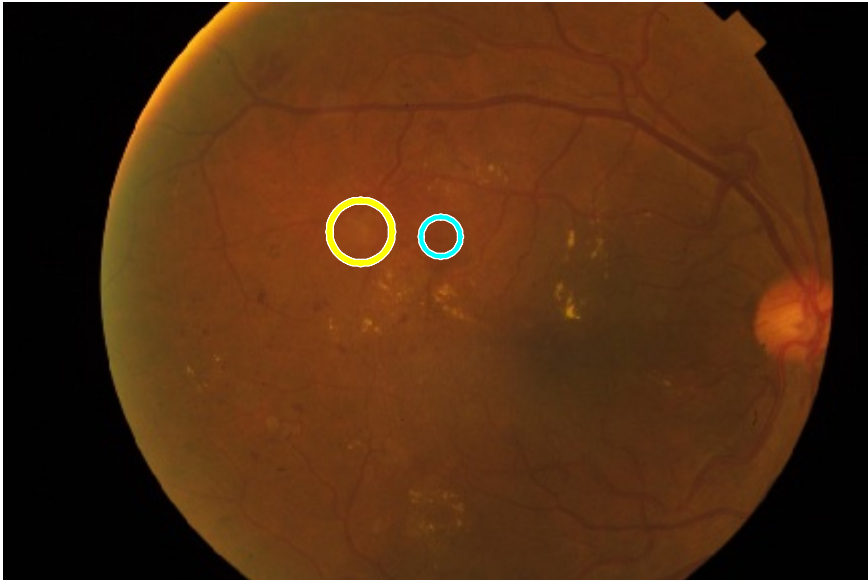


Enhanced (Quality: BORDERLINE, Score: 2/3)

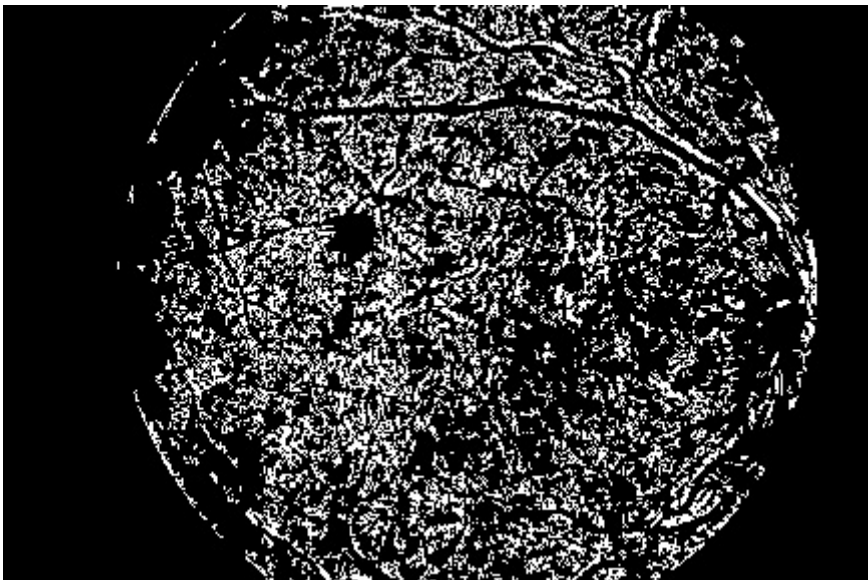


3. RETINAL SEGMENTATION & LANDMARK MAPS

Optic Disc (Yellow) & Fovea (Cyan)

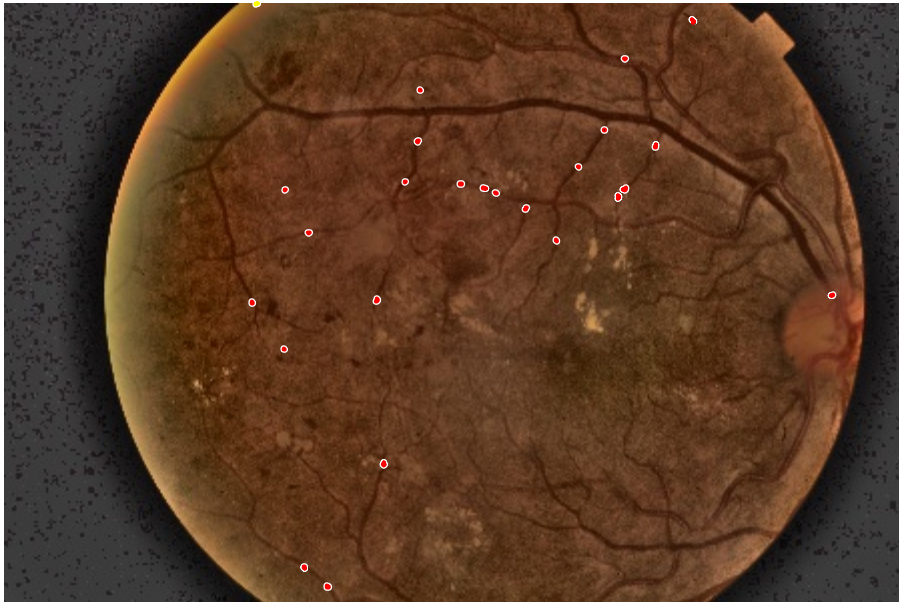


Segmented Vessel Network

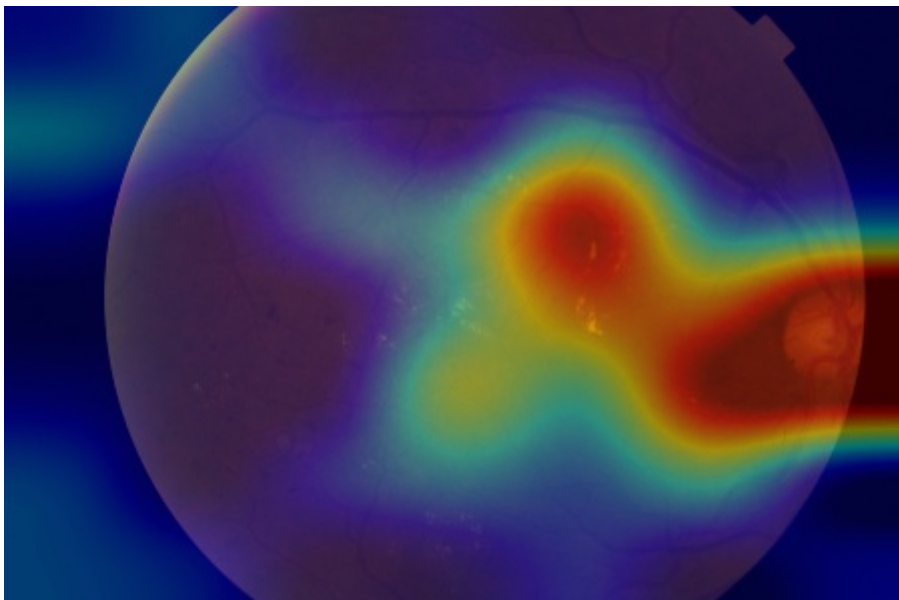


4. LESION DETECTION & EXPLAINABLE AI

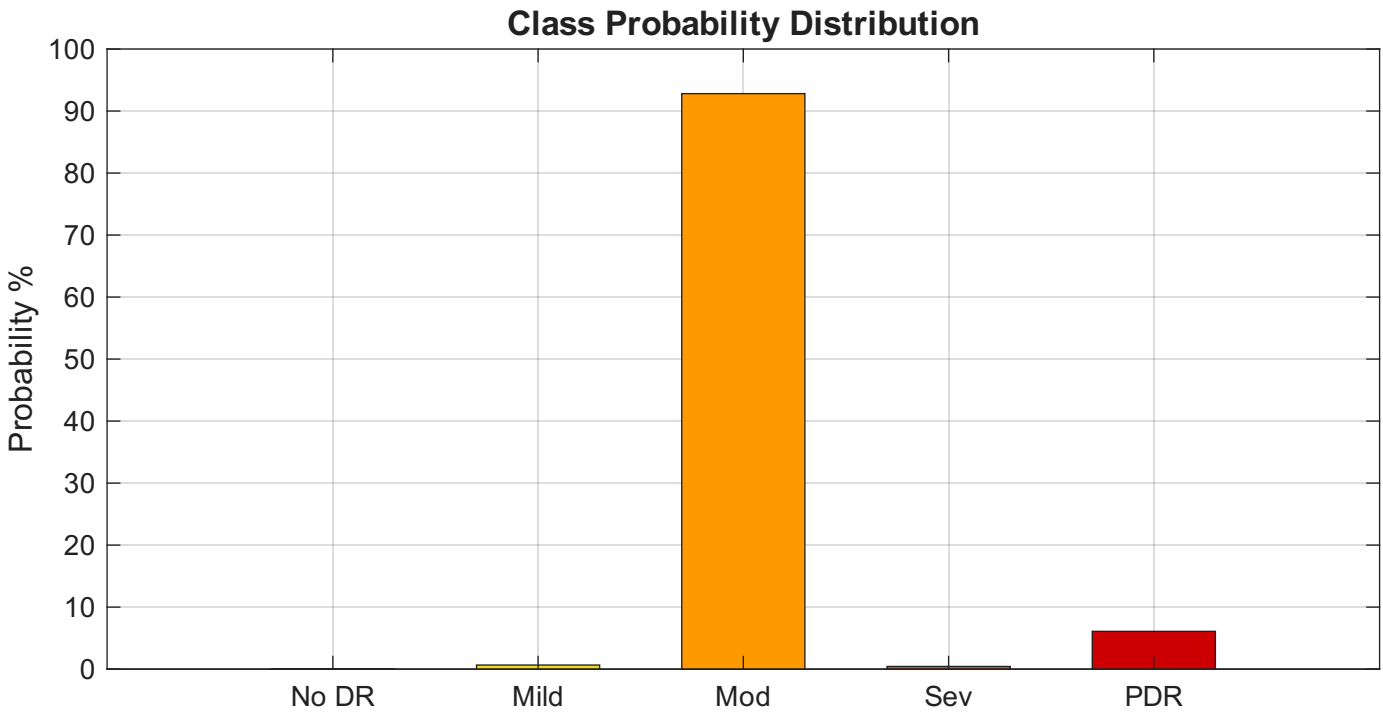
Detected Lesions – MA: 24 (Red) EX: 1 (Yellow) HE: 0 (Magenta)



Grad-CAM – Moderate NPDR (Confidence 92.8%)



5. CONFIDENCE ANALYSIS & TREATMENT RECOMMENDATION



Predicted Class: Moderate NPDR
Severity Level: 2 / 4
AI Confidence: 92.8%

Referral Needed: YES

ACTION: URGENT REFERRAL – Tele-consult within 7 days